ARTIFICIAL INTELLIGENCE • 2025

Large Language Models

From Transformers to ChatGPT and Beyond

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GPT-4 | Claude | Gemini | Llama | Transformers

Why Everyone Is Talking About AI

AI assistants have entered our daily work and personal lives

300M+ users in 60 days

ChatGPT became the fastest-growing app in history

ChatGPT
OpenAI • The pioneer

Copilot
Microsoft • Enterprise AI

Claude
Anthropic • Constitutional AI

Gemini
Google • Multimodal AI

DeepSeek
DeepSeek AI • Open source

Llama
Meta • Open weights

💡 **KEY TAKEAWAY:** LLMs are reshaping every industry - from healthcare to finance, education to software engineering.

The Evolution of Artificial Intelligence

From Alan Turing's vision to today's AI agents - a 75-year journey

- **1950** Alan Turing — Turing Test
- **1956** Dartmouth — AI is Born
- **1980s** Expert — Systems
- **2012** Deep — Learning
- **2017** Transformer — Revolution
- **2022** ChatGPT — Mass Adoption
- **2025+** AI — Agents

Before 2017: AI had limited language understanding. Models processed text sequentially - slow and forgetful.

After 2017 (Transformers): AI could finally understand context, meaning, and nuance at scale.

💡 **KEY TAKEAWAY:** The 2017 Transformer paper was the single biggest breakthrough in AI history. Everything changed.

What Is Language?

Language is patterns - and patterns can be learned by machines

Fill in the blank:

The capital of France is **Paris**

2 + 2 = **4**

The sky is **blue**

To be or not to **be**

Your brain instantly completes these!

Why language is learnable:

📊 **Statistical patterns**

Words co-occur in predictable ways

🧠 **Context matters**

Meaning depends on surrounding words

🔄 **Grammar rules**

Languages follow learnable structures

📚 **World knowledge**

Facts are embedded in how we write

💡 **KEY TAKEAWAY:** Language is fundamentally about predicting what comes next based on patterns learned from massive amounts of text.

What Is a Language Model?

At its core: a probability machine that predicts the next token

Definition: A Language Model assigns a probability to a sequence of words, and predicts the **next most likely token** given the previous tokens.

Example: Next token prediction

The	Eiffel	Tower	is	located	in	___

From Simple to Large:

Phone keyboard → Small model (~10K params)

Google search → Medium model (~millions)

GPT-4 → Large LLM (~1 Trillion params)

Most likely next tokens:

- Paris 92%
- France 5%
- Europe 2%
- Rome 1%

💡 **KEY TAKEAWAY:** A language model's magic: trained on trillions of words, it learns to predict the next token with remarkable accuracy.

How Autocomplete Became AI

The same core idea - scaled up by a trillion times

📱 Phone Keyboard

~2010

Suggests 3 words based on the last 1-2 words you typed. Local model.

Example: "I want to" → go / eat / be

~50K parameters

🔍 Google Search

~2015

Predicts your full query from first few characters using search history + patterns.

Example: "how to make..." → pasta / money / friends

~Millions of parameters

🤖 GPT-2

2019

First model that could generate coherent paragraphs. OpenAI initially restricted it!

Example: Full essay generation, code, stories

1.5B parameters

🧠 ChatGPT (GPT-4)

2022-23

Understands instructions, context, nuance. Engages in multi-turn conversations.

Example: Write code, analyze data, explain concepts

~1 Trillion parameters

💡 **KEY TAKEAWAY:** Same idea - predict next word - but scaled from 50K to 1 Trillion parameters. Scale changes everything.

The Problem Before Transformers

RNNs and LSTMs tried to solve language - but had fundamental limitations

❌ Recurrent Neural Networks (RNNs)

The — bank — of — the — river

Memory fades...

The | bank | of | the | river

← faded *fresh →*

- 🐌 Sequential - cannot parallelize
- ❓ Forgets long-term context
- 💀 Vanishing gradient problem
- ⏱️ Very slow to train

⚠️ LSTMs - An Improvement, But...

Long Short-Term Memory Networks

- ✅ Better memory than RNNs
- ✅ Handles longer sequences
- ❌ Still sequential (slow)
- ❌ Struggles with 1000+ words
- ❌ Hard to parallelize on GPUs
- ❌ Complex to train

💡 **KEY TAKEAWAY:** RNNs and LSTMs had fundamental architectural limits. The AI field needed a completely new approach.

2017: The Breakthrough That Changed Everything

Google Brain • June 12, 2017 • arXiv:1706.03762

Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar,
Jakob Uszkoreit, Llion Jones, Aidan N. Gomez,
Łukasz Kaiser, Illia Polosukhin

Google Brain & Google Research

Citations: 100,000+

Published: June 2017

🔬 *arXiv:1706.03762*

Why this paper changed everything:

1. **No More Sequential Processing**

   Processes ALL words simultaneously in parallel. Training time dropped from months to days.

2. **Attention Mechanism**

   Every word can directly attend to every other word. No more memory fading.

3. **Infinitely Scalable**

   More data + more compute = better results. This drove the LLM arms race.

💡 **KEY TAKEAWAY:** "Attention Is All You Need" is one of the most cited papers in history. It gave birth to BERT, GPT, and every modern LLM.

The Transformer Architecture

High-level architecture - the foundation of every modern LLM

Input Text
"What is AI?"

Tokenization
Split into tokens

Embeddings
Convert to vectors

Attention Layers
Find relationships

Feed Forward Network
Transform features

Output
Predicted tokens

Encoder vs Decoder:
Some models encode input (BERT), some decode output (GPT). ChatGPT uses a decoder-only architecture.

Layers Stack Up:
Multiple transformer blocks are stacked (12 to 96+ layers). Each layer learns progressively more complex patterns.

Multi-Head Attention:
Multiple attention heads run in parallel, each focusing on different aspects of meaning and relationships.

Scale = Power:
More layers + more parameters + more data = dramatically better performance. This is the scaling law.

💡 **KEY TAKEAWAY:** Input text → Tokens → Vectors → Attention layers → Feed Forward → Output. This pipeline powers every ChatGPT response.

What Are Tokens?

LLMs don't read words - they read tokens (word pieces)

"Tokenization is fascinating and fundamental to language models"

↓ *Tokenized (13 tokens):*

"Token" "ization" " is" " fas" "cin" "ating" " and" " fund" "amental" " to" " language" " models"

Whole Words

"cat", "dog", " and"

Sub-words

"Token" + "ization"

Characters

"a", "b", "c"

Special

[START], [END], [PAD]

📏 Rule of thumb: ~1 token = ¾ of a word | 100 tokens ≈ 75 words

💡 **KEY TAKEAWAY:** GPT-4's vocabulary has ~100,000 tokens. Every LLM response is built one token at a time.

What Are Embeddings?

Converting words into numbers - the language machines understand

Words → Numbers (Vectors)

King → `[0.82, 0.31, 0.77, 0.45...]`

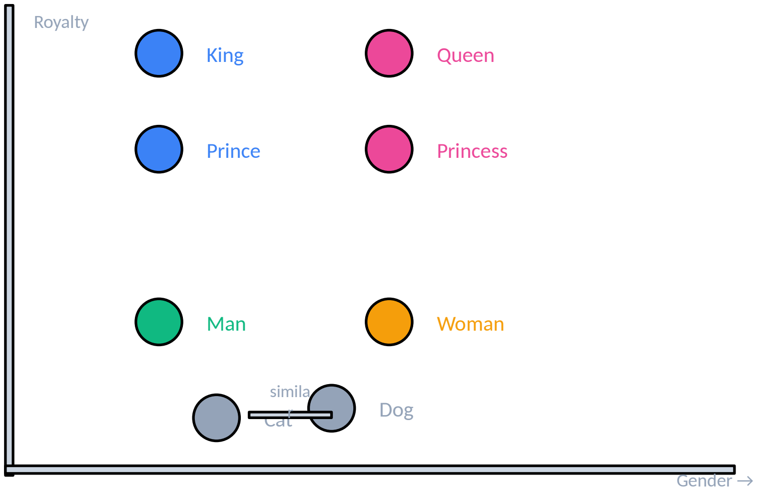
Queen → `[0.79, 0.70, 0.76, 0.48...]`

Man → `[0.82, 0.30, 0.45, 0.22...]`

Woman → `[0.80, 0.68, 0.44, 0.24...]`

King − Man + Woman ≈ Queen (vector math works!)

Semantic Space (conceptual 2D view)



💡 **KEY TAKEAWAY:** Embeddings place words with similar meanings close together in mathematical space - enabling semantic understanding.

What Is the Attention Mechanism?

The revolutionary idea: every word can look at every other word simultaneously

Human Reading Analogy: When reading "The **animal** didn't cross the street because **it** was too tired" - what does "it" refer to?

The	**animal**	didn't	cross	the	street	because	**it**	was	tired

← Attention connection ("it" refers to "animal")

Q - Query

What am I looking for?

Token "it" asks: "who/what do I refer to?"

K - Key

What can I offer?

Token "animal" says: "I'm a noun that can be referred to"

V - Value

What information do I carry?

"animal" passes its meaning to "it"

💡 **KEY TAKEAWAY:** Attention = Each word dynamically weighs how relevant every other word is to understanding its own meaning.

Softmax: From Scores to Probabilities

Softmax converts raw attention scores into probabilities that sum to 1.0

Raw Attention Scores

(Q · K dot product)

The	0.3
animal	8.7
didn't	1.2
cross	0.8
street	2.4

→

Softmax

After Softmax (Probabilities)

(sum = 1.0)

The	1%
animal	84%
didn't	4%
cross	2%
street	9%

✓ 1% + 84% + 4% + 2% + 9% = 100% (always sums to 100%)

💡 **KEY TAKEAWAY:** Softmax ensures attention weights always sum to 1.0 - making them true probabilities that can be compared and used.

Multi-Head Attention

Not just one attention - many heads, each learning different relationships

Head 1
Focus: Syntax

Who is doing what to whom?

e.g. subject-verb

Head 2
Focus: Coreference

What does this pronoun refer to?

e.g. it → animal

Head 3
Focus: Semantic

What concepts are related?

e.g. bank → river

Head 4
Focus: Position

What is near vs far?

e.g. adjacent words

Head 5
Focus: Negation

What is being negated?

e.g. didn't → cross

Head 6
Focus: Entities

What are the nouns?

e.g. noun phrases

↓ All 6 heads concatenated → projected through linear layer → richer representation

💡 **KEY TAKEAWAY:** GPT-4 uses 96 attention heads per layer × 96 layers. Each head learns a unique type of linguistic or factual relationship.

How LLMs Are Trained

Three phases: Pre-training → Fine-tuning → RLHF alignment

Phase 1

Pre-training

Train on trillions of words from books, web, Wikipedia, code. Task: predict the next token.

Data: ~15 Trillion tokens

Cost: ~$100M+ compute

Time: Months on thousands of GPUs

- Books & literature
- Wikipedia
- GitHub code
- Scientific papers
- Web pages (filtered)

Phase 2

Fine-tuning (SFT)

Train on high-quality human demonstrations. Teaches the model to follow instructions.

Data: ~100K examples

Cost: ~$1M compute

Time: Days

- Human-written Q&A
- Instruction following
- Helpful responses
- Task demonstrations

Phase 3

RLHF Alignment

Reinforcement Learning from Human Feedback. Humans rank responses; model learns to prefer better answers.

Data: Human preference data

Cost: ~$10M (labeler time)

Time: Weeks

- Human raters compare pairs
- Reward model trained
- Policy optimized (PPO)
- Safety filters applied

💡 **KEY TAKEAWAY:** Training ChatGPT cost ~$100M+ in compute. But once trained, inference costs pennies per query - scale economics work.

Evolution of Modern Language Models

From BERT to GPT-4o - seven years of exponential progress

2018
BERT
Google • 340M
Bidirectional understanding

2019
GPT-2
OpenAI • 1.5B
Generative writing shocked the world

2020
GPT-3
OpenAI • 175B
Few-shot learning emergence

2022
ChatGPT
OpenAI • 175B+
RLHF. 100M users in 60 days

2023
GPT-4
OpenAI • ~1T
Multimodal. Expert-level reasoning

2024
Gemini
Google • Undisclosed
Native multimodal. 1M context

💡 **KEY TAKEAWAY:** Each generation was not an incremental update - it was a qualitative leap in capability. Parameters grew 3,000x in 6 years - scale = intelligence.

How ChatGPT Generates Responses

Step-by-step inference pipeline - what happens in milliseconds

1. **User Input** — *"What is photosynthesis?"*
2. ✂️ **Tokenize** — *Split into 4 tokens*
3. **Embed** — *Each token → 12,288D vector*
4. ⚡ **Transform** — *96 layers of attention*
5. **Probability** — *Scores for ~100K tokens*
6. **Sample** — *Pick next token*
7. **Repeat** — *Until [END] token*

Auto-regressive Generation (one token at a time):

Photo	synthesis	is	the	process	by	which	**plants...**

💡 **KEY TAKEAWAY:** Every word you see from ChatGPT is generated one token at a time. A 500-word response = ~650 individual sampling decisions.

Temperature, Top-K & Top-P Sampling

Controlling creativity vs accuracy in LLM responses

🌡️ Temperature

Low Temp (0.1)

Paris 95%

Lyon 3%

Nice 1%

Rome 0.5%

🎯 Predictable
✅ Factual tasks (Q&A, analysis)

High Temp (1.5)

Paris 35%

Lyon 28%

Nice 22%

Rome 12%

🎨 Creative
✅ Writing, brainstorm (poems, stories)

🔝 Top-K Sampling

Only consider the top K tokens by probability. K=50 means: only pick from the 50 most likely tokens, ignoring all others.

✅ Prevents rare/random tokens from being selected

📊 Top-P (Nucleus) Sampling

Consider only the smallest set of tokens whose cumulative probability exceeds P. P=0.9 means: sample from tokens that account for 90% of the probability mass.

✅ Adapts to situations - wider when uncertain, narrower when confident

💡 **KEY TAKEAWAY:** ChatGPT typically uses Temperature ≈ 1.0, Top-P ≈ 0.95. Coding assistants use lower temperatures for determinism.

Do LLMs Have Memory?

No - but context windows simulate memory. Here's how it works.

NO

LLMs have zero persistent memory between sessions

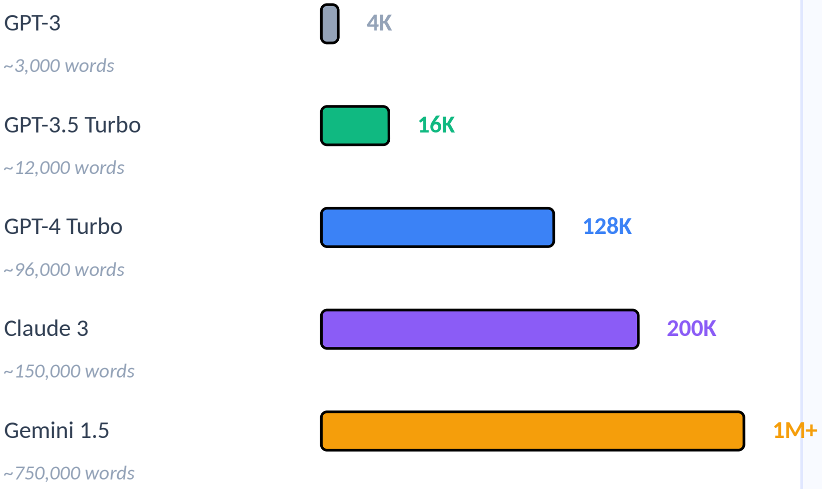
What IS Context?

- 📋 System prompt (instructions)
- 💬 Conversation history (all turns)
- 📄 Uploaded documents
- 🔍 Retrieved documents (RAG)

💡 Analogy: Context = everything you see on your desk right now (not what's in storage)

Context Window Sizes

(= working memory size)



💡 **KEY TAKEAWAY:** ChatGPT 'remembers' your conversation because the entire chat history is sent to the model with every new message.

SECTION 10 • MEMORY & CONTEXT

RAG: Retrieval Augmented Generation

Connecting LLMs to your enterprise knowledge base

📚 **Knowledge Base (Vector Database)**

PDFs | Contracts | Wiki | Emails | Reports | Code

User Query → Vector Search → Retrieve Chunks → Build Prompt → LLM Generates

🔄 Always Current
Retrieves fresh data from your knowledge base - no retraining needed

📎 Source Citations
Can cite exact documents it retrieved - fully auditable

🔒 Private Knowledge
Your proprietary data never leaves your control

💰 Cost Efficient
No need to fine-tune expensive models for each update

💡 **KEY TAKEAWAY:** RAG is how enterprises deploy LLMs with their own data - without retraining models. It's the most common enterprise AI pattern.

Limitations of Large Language Models

Understanding limitations is essential for responsible AI deployment

Context Window Limits

MED

Cannot process documents longer than the context window. Instructions at the start of a long document may be 'forgotten'.

e.g. Analysis of a 1000-page book requires chunking

Knowledge Cutoff

MED

Training data has a fixed end date. The model has no knowledge of events after this date unless given new information.

e.g. GPT-4 doesn't know about events after April 2024

Bias & Stereotypes

HIGH

Training data reflects human biases. Models can produce discriminatory or stereotyped responses.

e.g. Gender, racial, cultural biases from training data

Cost & Latency

LOW

Large models are expensive to run. Real-time applications may face latency constraints.

e.g. GPT-4 API: ~$0.01-0.03 per 1K tokens

💡 **KEY TAKEAWAY:** LLMs are powerful tools, not infallible oracles. Always verify important outputs. Design systems with human oversight.

The Enterprise AI Landscape

Key players shaping the AI industry - who does what

OpenAI
Pioneer / Platform

GPT-4, ChatGPT, o1, Sora

First to mass market. API leader. Backed by Microsoft ($13B).

Anthropic
Safety-first AI

Claude 3 (Haiku, Sonnet, Opus)

Founded by ex-OpenAI. Constitutional AI. Strong enterprise focus.

Google DeepMind
Research + Scale

Gemini Pro/Ultra, Imagen, PaLM

Huge research output. Integration across Google products.

Microsoft
Enterprise Integrator

Copilot, Azure OpenAI, Bing AI

$13B invested in OpenAI. AI into every Office product.

Meta AI
Open Source Leader

Llama 3, Code Llama, SAM

Open-weights models democratizing AI development globally.

NVIDIA
AI Infrastructure

H100, A100, CUDA, NIM

The picks-and-shovels of the AI gold rush. $3T valuation.

💡 **KEY TAKEAWAY:** The AI landscape is an ecosystem: NVIDIA provides compute, model providers build LLMs, Microsoft/Google integrate into enterprise tools.

Open Source vs Proprietary AI Models

Two paradigms - each with compelling advantages for enterprise

🔒 Proprietary Models

OpenAI • Anthropic • Google

Advantages:

✅ Best-in-class performance

✅ Fully managed - no infra needed

✅ Constantly updated

✅ Enterprise support & SLA

✅ Built-in safety filters

Disadvantages:

❌ Ongoing API costs

❌ Data leaves your environment

❌ Vendor lock-in risk

VS

🌐 Open Source Models

Llama • Mistral • DeepSeek • Gemma

Advantages:

✅ Full data privacy & control

✅ Deploy on your own infra

✅ Fine-tune for your domain

✅ No per-query API costs

✅ Community innovation

Disadvantages:

❌ Requires ML engineering

❌ GPU infrastructure cost

❌ May lag frontier models

💡 **KEY TAKEAWAY:** Most enterprises use BOTH: proprietary APIs for general tasks, open source models for sensitive/regulated data where privacy is critical.

AI Governance, Security & Responsible AI

Enterprise AI must be secure, compliant, and aligned with human values

🔐 AI Security

Prompt Injection

Malicious inputs hijack model behavior via crafted prompts

Data Leakage

Sensitive training data or context exposed in outputs

Jailbreaks

Bypassing safety filters through clever prompt engineering

Model Theft

Extracting model weights through query patterns

⚖️ Responsible AI

Fairness

AI decisions must not discriminate against protected groups

Transparency

Users must know when they are interacting with AI

Explainability

Decisions affecting people must be explainable

Human Oversight

Critical decisions require human review and approval

📋 Enterprise Governance

AI Policy

Acceptable use policy covering what AI can/cannot be used for

Risk Management

Assess and mitigate AI-specific risks before deployment

Compliance

GDPR, SOC2, ISO 42001 AI management systems

Auditability

Log AI decisions and maintain audit trails

💡 **KEY TAKEAWAY:** AI governance is not optional - EU AI Act applies globally to any AI serving EU citizens. Build compliance from day one.

The Future of Artificial Intelligence

From LLMs to AI agents, reasoning models, and physical AI

AI Agents
Now

AI that can plan, reason, use tools (web search, code execution, APIs) and complete complex multi-step tasks autonomously.

→ Auto-code reviews, research automation, workflow agents

Reasoning Models
Now

Models trained specifically to 'think before answering' - dramatically better at math, science, and complex logic.

→ OpenAI o1, DeepSeek R1, Google Gemini Thinking

Multimodal AI
Now

Seamlessly process and generate text, images, audio, video, and code. True understanding across all information types.

→ GPT-4o, Gemini Ultra, Claude Vision

Multi-Agent Systems
2025

Multiple specialized AI agents collaborating - one researches, one codes, one reviews, one deploys. Automated pipelines.

→ AutoGen, CrewAI, LangGraph, Devin

World Models
2025

AI that builds internal simulations of the physical world - enabling prediction, planning, and physical reasoning.

→ OpenAI Sora, Meta V-JEPA, Google Genie

Physical AI / Robotics
2026

LLMs as the 'brain' of robots. Natural language to physical action. Generalist robots that learn from observation.

→ Tesla Optimus, Figure AI, Google RT-2

💡 **KEY TAKEAWAY:** We are in the early innings. The transition from AI as a tool to AI as an autonomous collaborator will reshape every industry.

“

The Future Belongs to
Those Who Understand AI

AI fluency is the most valuable skill for the next decade.

🎯 Ex [illegible] Copilot

Thank You

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🙋 **Questions & Discussion**

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References & Further Reading

Attention Is All You Need

Vaswani et al. (2017) - Google Brain / NeurIPS
arxiv.org/abs/1706.03762

BERT: Pre-training of Deep Bidirectional Transformers

Devlin et al. (2018) - Google AI Language
arxiv.org/abs/1810.04805

Language Models are Unsupervised Multitask Learners (GPT-2)

Radford et al. (2019) - OpenAI
openai.com/research/language-unsupervised

Language Models are Few-Shot Learners (GPT-3)

Brown et al. (2020) - OpenAI / NeurIPS
arxiv.org/abs/2005.14165

Training language models to follow instructions (InstructGPT)

Ouyang et al. (2022) - OpenAI / NeurIPS
arxiv.org/abs/2203.02155

GPT-4 Technical Report

OpenAI (2023) - OpenAI
arxiv.org/abs/2303.08774

Gemini: A Family of Highly Capable Multimodal Models

Gemini Team, Google (2023) - Google DeepMind
arxiv.org/abs/2312.11805

Llama 2: Open Foundation and Fine-Tuned Chat Models

Touvron et al., Meta (2023) - Meta AI
arxiv.org/abs/2307.09288

Further Learning:

• Andrej Karpathy: "Let's build GPT from scratch" (YouTube) • fast.ai: Practical Deep Learning • DeepLearning.AI: Generative AI courses • Anthropic's Interpretability Research